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MIGHTY TRONICS®

PUMP INTERFACE MASTER DRAWING – REV A

Version & Revision
V 1.0

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LIST OF CONTENTS

1.	History of Version	2
2.	identification and drawing authority	3
3.	Board outline and reference view	3
4.	Layer structure and ordered fabrication options	5
5.	Drill schedule	5
6.	General fabrication notes	6
7.	Engineering release checks.....	6
8.	Design Documentation and KiCad Exports	7
8.1.	Preserve the manufactured revision.....	7
8.2.	Required documentation exports	7
8.3.	Schematic	8
8.4.	PCB overview	8
8.5.	Copper layer	9
8.6.	Fabrication and outline	11
8.7.	Drill map and drill report.....	11
9.	Useful links.....	12

1. HISTORY OF VERSION

Date	Version	Amendments
27-09-2026	V1.0	Document initiation

2. IDENTIFICATION AND DRAWING AUTHORITY

Item	Definition
Board / design	MT Drive — Pump Interface; four-layer rigid PCB
Fabrication package	Pump Interface-GerberExcellon_Y2 (JLCPCB product detail screenshot)
Manufacturer / lot	JLCPCB / 5 individual PCBs; one design
Source CAD	Pump Interface.kicad_pcb; native board file not supplied for this revision
Data precedence	Approved fabrication data and order settings govern geometry and process. Resolve any conflict with this drawing before fabrication. Do not scale the illustration.
Drawing status	Documentation draft for the existing prototype order; not released as a new fabrication authority

3. BOARD OUTLINE AND REFERENCE VIEW

Property	Specified value / source
Overall dimensions	157.5 × 111.8 mm, as displayed in the JLCPCB order detail
Outline tolerance	±0.2 mm (Regular), as selected in the JLCPCB order
Board thickness	1.6 mm nominal; manufacturing tolerance per JLCPCB order/process
Board construction	Four-layer FR-4, material option TG135; dielectric stack-up not separately specified
Mounting holes	4 × Ø3.20 mm NPTH, subject to Excellon and Edge.Cuts verification

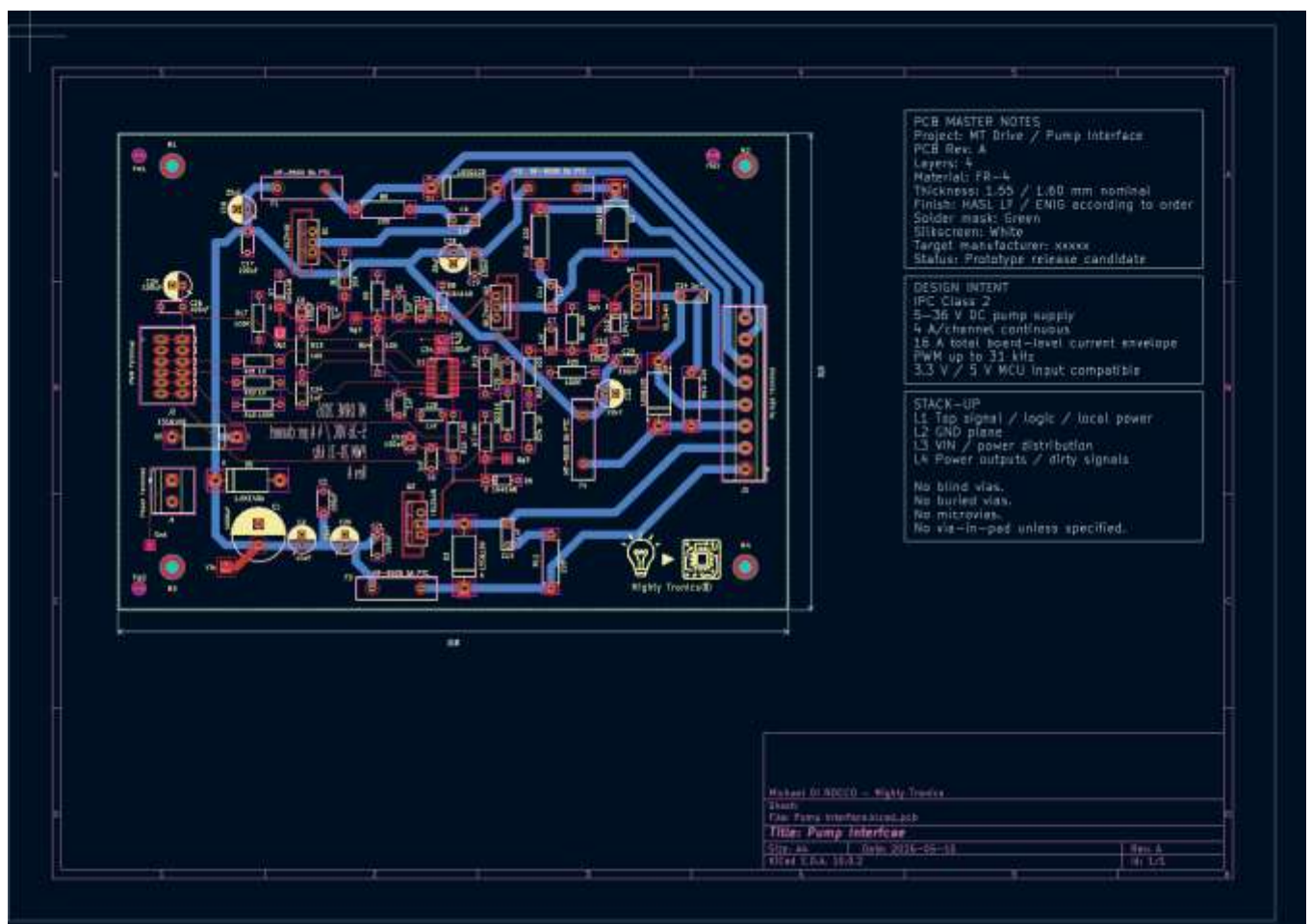


Figure 1 - Supplied KiCad drawing screenshot, shown for reference only. Some embedded notes reflect earlier assumptions; the verified order settings in Sections 3 and 4 take precedence. The drawing image must not be used to derive dimensions.

4. LAYER STRUCTURE AND ORDERED FABRICATION OPTIONS

Layer / feature	Function and order setting
L1 / F.Cu	Top copper, nominal 1 oz (approximately 35 µm copper weight equivalent); signals, logic and local power
L2 / In1.Cu	Inner copper, nominal 2 oz (approximately 70 µm); GND plane
L3 / In2.Cu	Inner copper, nominal 2 oz (approximately 70 µm); VIN / power distribution
L4 / B.Cu	Bottom copper, nominal 1 oz (approximately 35 µm); power outputs and other routed nets
Layer sequence / dielectrics	Four layers ordered; "No requirement" for a specified stack-up. Confirm actual dielectric construction with the manufacturer if controlled impedance or dielectric spacing is required.
Surface finish	HASL with lead (SnPb), as ordered. Do not label this finish lead-free.
Solder mask / legend	Green solder mask; white ink-jet silkscreen
Via treatment	Covered vias ("Bouché" in the translated order detail); confirm the exact JLCPCB process meaning from production data if plugging/filling is required
Electrical test	Full flying-probe electrical test
Appearance quality	JLCPCB "IPC Class 2 Standard" option; not independent evidence of full IPC-6012 certification
Other selections	No castellations, blind holes, edge plating, countersinks, gold fingers, UL marking or inspection report ordered

Copper weights describe the manufacturer order selections. The approximate 35/70 µm conversions are nominal copper-weight equivalents, not measured final thicknesses or guaranteed finished copper tolerances. External plating may affect local finished thickness.

5. DRILL SCHEDULE

Tool	Hole type	Nominal Ø	Count	Plated
T1	PTH	0.40	4	Yes
T2	PTH	0.65	8	Yes
T3	PTH	0.80	84	Yes
T4	PTH	0.90	12	Yes
T5	PTH	1.00	10	Yes
T6	PTH	1.20	12	Yes
T7	PTH	1.25	2	Yes
T8	PTH	1.30	12	Yes
T9	PTH	1.40	20	Yes
T10	NPTH	3.20	4	No

Schedule total: 164 plated through holes and 4 non-plated through holes, 168 holes overall. Verify the schedule against the final Excellon files. Confirm whether the workbook

numbers represent required finished hole diameters or nominal drill tool sizes; these conventions must not be mixed.

6. GENERAL FABRICATION NOTES

- Fabricate to the approved four-copper-layer Gerber data, plated and non-plated Excellon files, Edge.Cuts profile, and manufacturer order settings. Query any discrepancy before manufacture.
- The manufacturer order specifies 1 oz outer and 2 oz inner copper. The 2 oz selection applies to BOTH L2 and L3, not only to the power plane.
- Provide plated through holes for T1–T9 and non-plated mounting holes for T10, after drill-file reconciliation. No blind or buried vias or microvias are specified.
- Apply solder mask and silkscreen as defined by the fabrication data; maintain legible reference markings and keep ink off exposed solderable pads.
- The PCB is supplied as individual boards, without panelization. Component placement, polarity and soldering acceptance belong to a separate assembly drawing and inspection plan.
- Do not infer electrical current rating from copper weight alone. The product positioning is 2 A continuous per channel, 8 A total guaranteed; operation up to 3 A per channel may be possible but is not guaranteed. This is a product performance statement requiring validation, not a bare-board acceptance criterion.

7. ENGINEERING RELEASE CHECKS

ID	Required disposition before releasing a new fabrication revision
R-01	Compare native .kicad_pcb outline, actual layer assignments, copper pours, and dimension annotations with the 157.5 × 111.8 mm ordered dimensions and the final Gerbers.
R-02	Reconcile every Excellon tool, count and PTH/NPTH classification against Section 4. Confirm finished-hole versus drill-size convention.
R-03	Check MOSFET TO-220 footprint and real lead geometry: the workbook lists a 1.20 mm hole while its calculated lead envelope reaches 1.43 mm. The workbook "OK" flag does not resolve this discrepancy.
R-04	Resolve overlapping/ambiguous component reference groupings in the hole-to-component table, including C3. Check against the BOM and PCB footprints.
R-05	Run PCB DRC and review the plotted Gerbers and Excellon in GerbView and in the JLCPCB preview. Record exceptions and archive the approved upload package.
R-06	If RoHS/lead-free assembly is intended, resolve the ordered HASL-with-lead finish and document an appropriate surface finish for a future revision.

8. DESIGN DOCUMENTATION AND KICAD EXPORTS

The purpose of the following exports is to document the design and support test results, review, and future revisions.

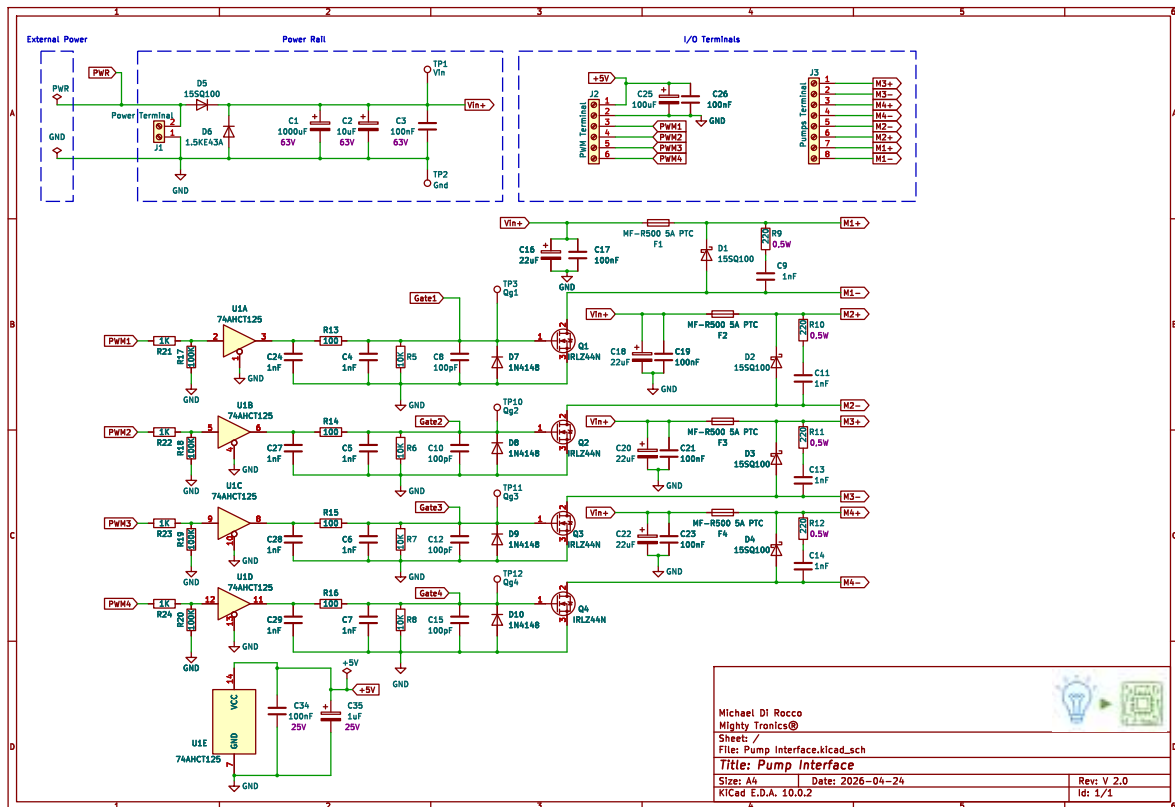
8.1. PRESERVE THE MANUFACTURED REVISION

Archive the exact Pump Interface-GerberExcellon_Y2 package submitted to JLCPCB, together with the corresponding native KiCad project files and the JLCPCB order details. Identify this package as the manufactured revision. If the KiCad board has since been modified, label the current files separately; newly generated exports must not be presented as the files used to manufacture this lot.

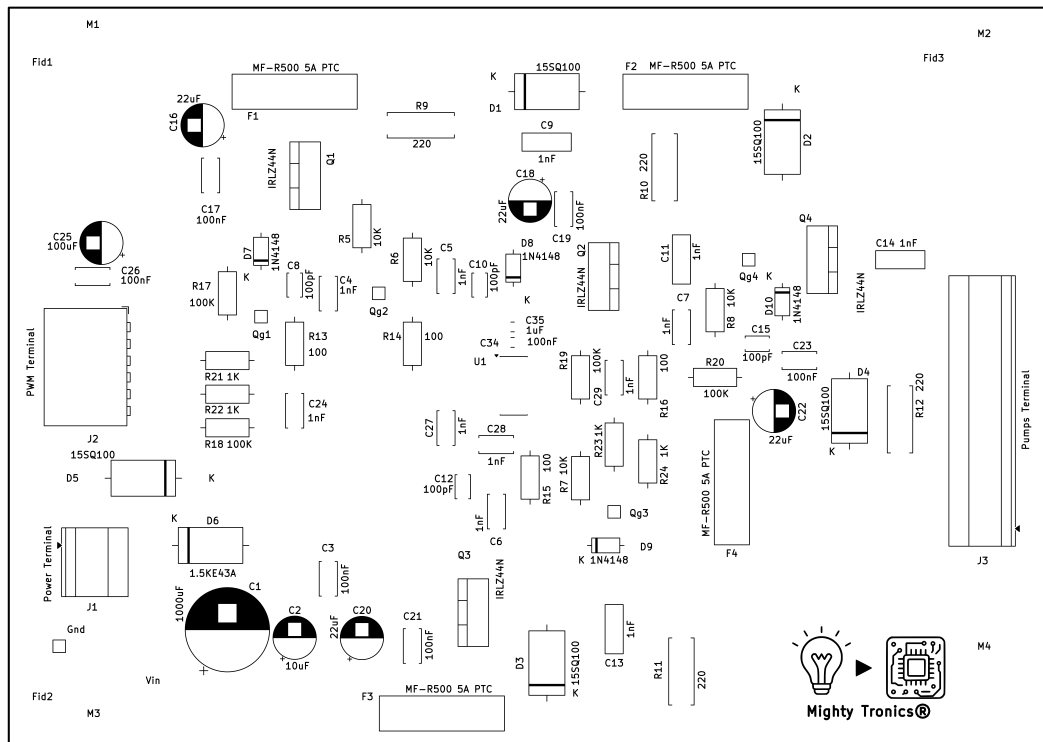
8.2. REQUIRED DOCUMENTATION EXPORTS

Document or data	Purpose
Schematic PDF	Records circuit connections, component references, values, and interface names
PCB overview PDF	Provides a readable overall view for the design and test file
Copper layer PDFs	Records routing and the two internal planes for engineering review
Fabrication and outline PDF	Records the board profile, mounting features, and fabrication annotations
Drill map and drill report	Supports verification of hole locations, tool diameters, and PTH/NPTH separation against the hole-control spreadsheet
DRC report	Records the design-rule status of the archived KiCad revision; document any accepted exceptions

8.3.SCHEMATIC



8.4.PCB OVERVIEW



8.5.COPPER LAYER

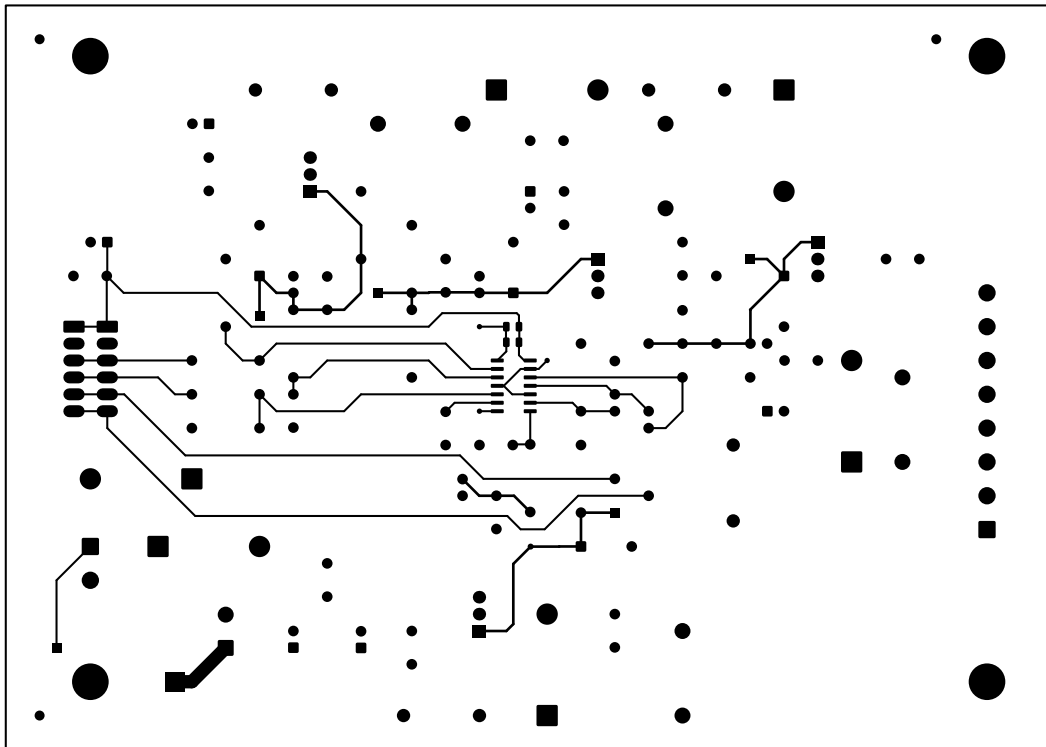


Figure 2 - L1 copper layer (signal command)

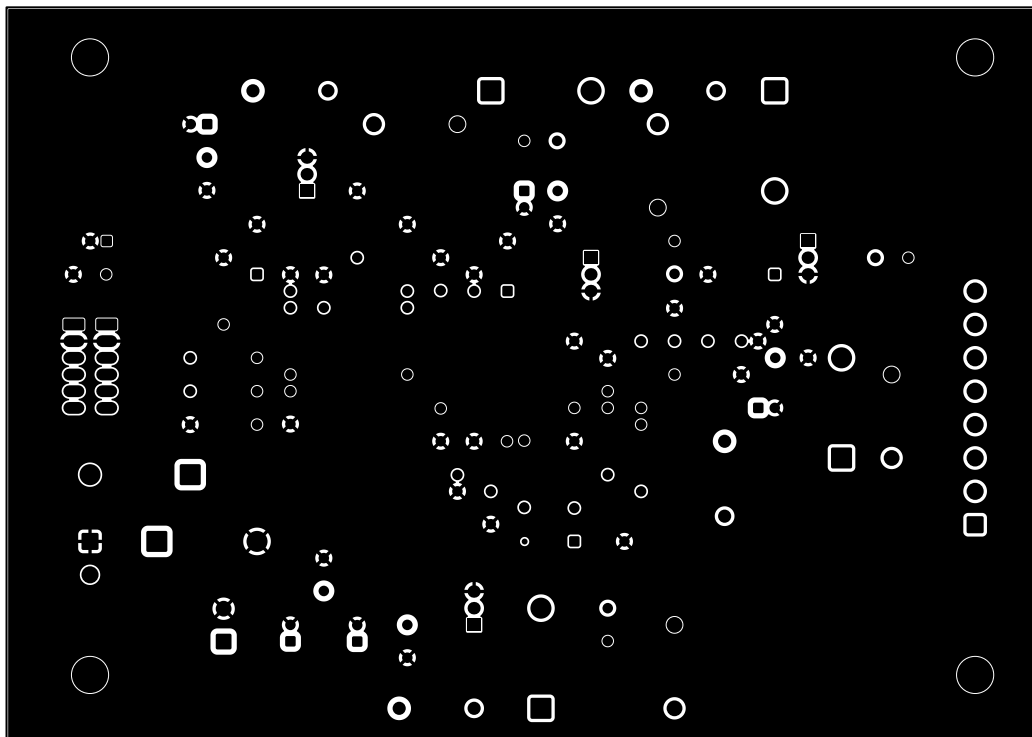


Figure 3 - L2 copper layer (Ground plane)

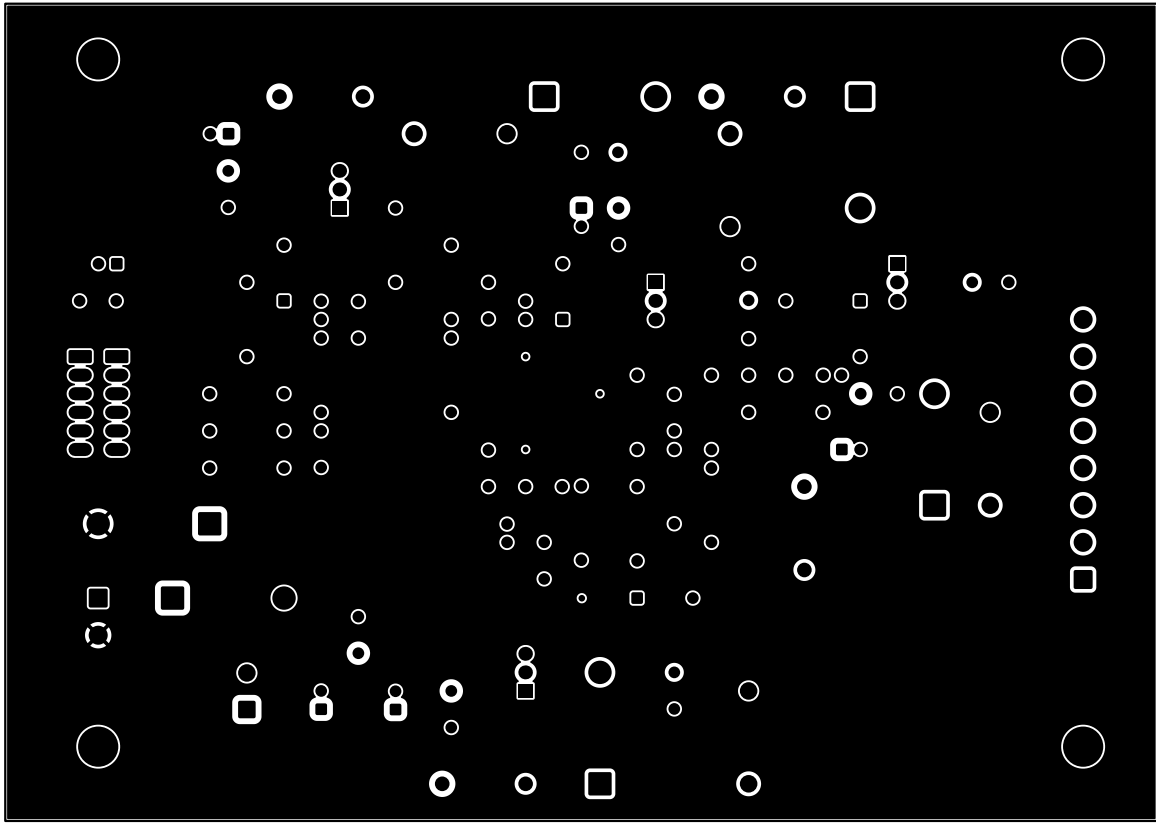


Figure 4 - L3 copper layer (Power supply plane)

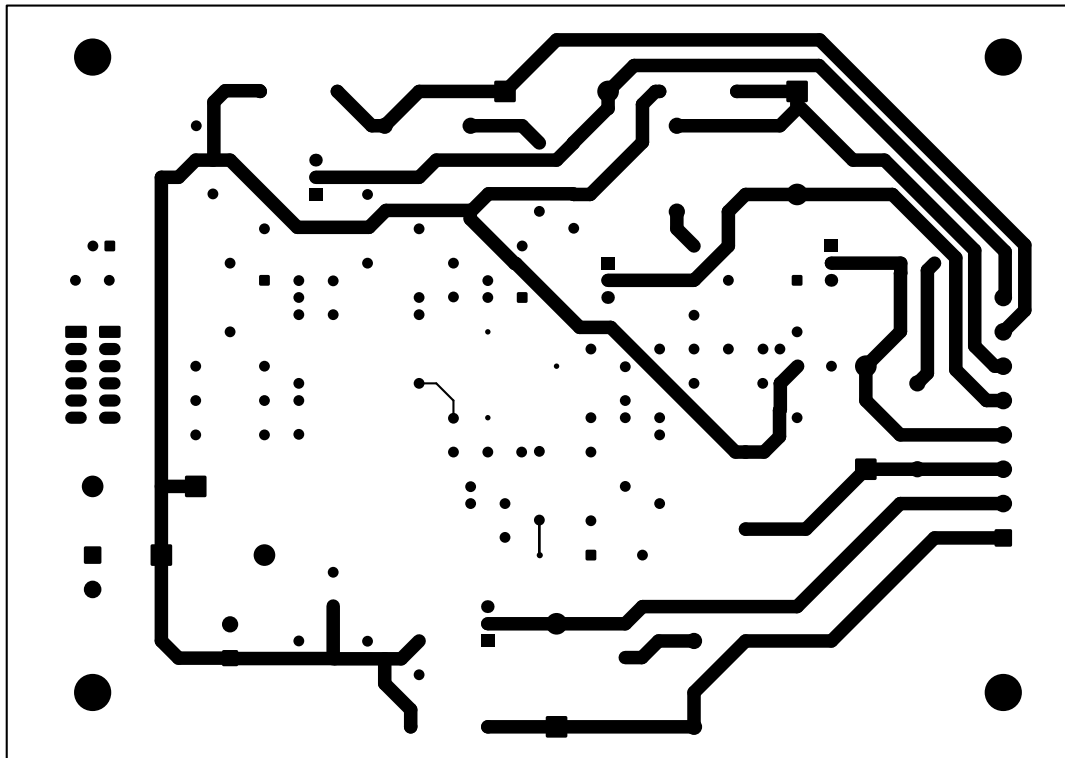
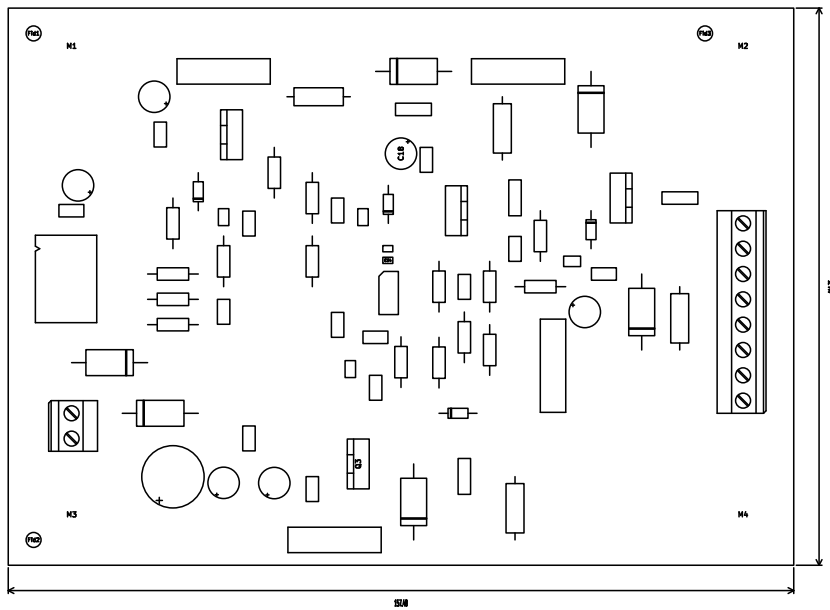


Figure 5 - L4 copper layer (dirty signals)

8.6.FABRICATION AND OUTLINE

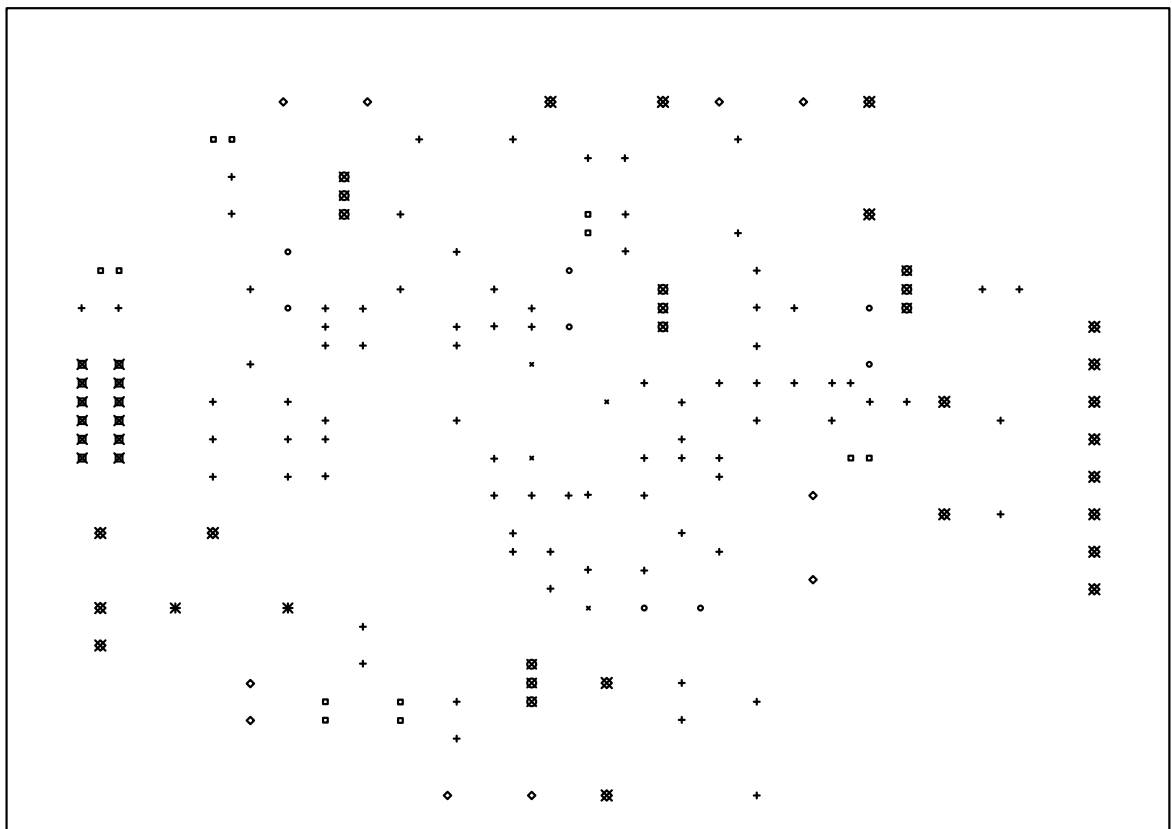


PCB MASTER NOTES
 Project: MT Drive / Pump Interface
 PCB Rev: A
 Layers: 4
 Material: FR-4
 Thickness: 1.55 / 1.60 mm nominal
 Finish: HASL LF / ENIG according to order
 Solder mask: Green
 Silkscreen: White
 Target manufacturer: xxxxx
 Status: Prototype release candidate

DESIGN INTENT
 IPC Class 2
 5-36 V DC pump supply
 4 A/channel continuous
 16 A total board-level current envelope
 PWM up to 31 kHz
 3.3 V / 5 V MCU input compatible

STACK-UP
 L1 Top signal / logic / local power
 L2 GND plane
 L3 VIN / power distribution
 L4 Power outputs / dirty signals
 No blind vias.
 No buried vias.
 No microvias.
 No via-in-pad unless specified.

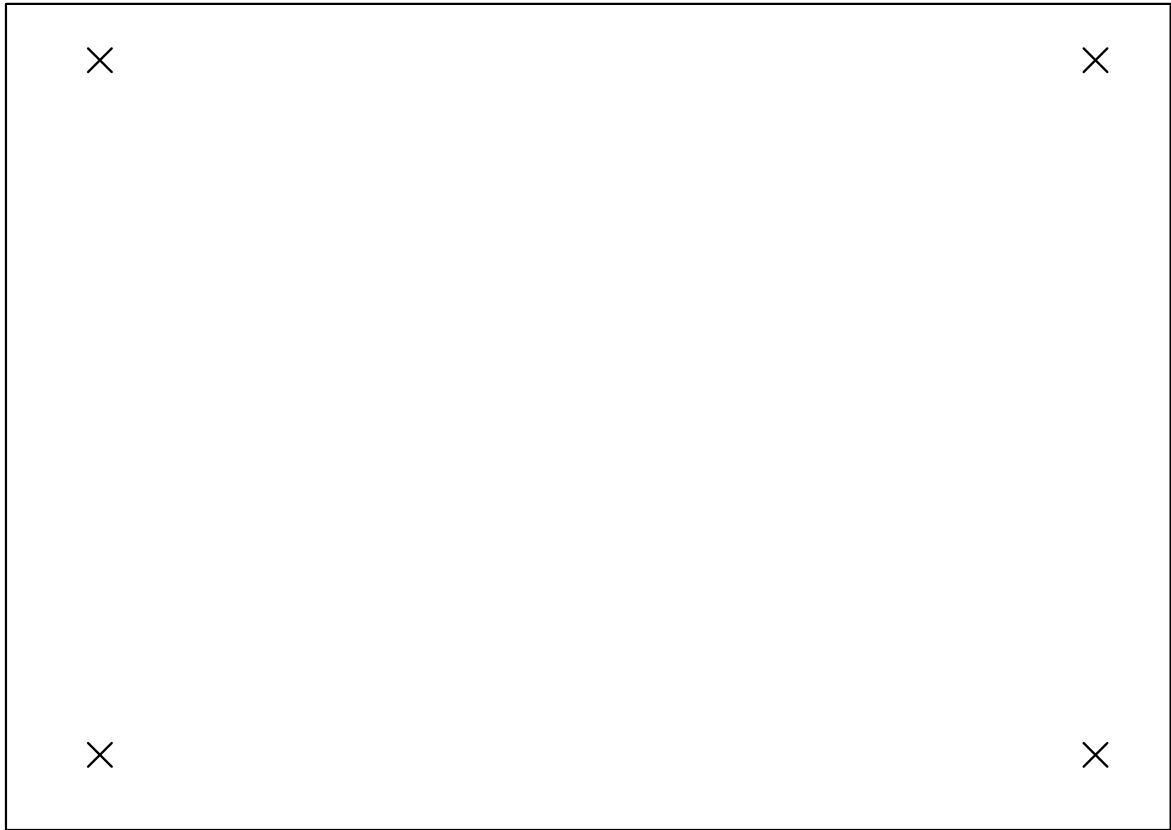
8.7.DRILL MAP AND DRILL REPORT



Drill Map:

- * 0.400mm / 0.0157" (4 holes)
- o 0.650mm / 0.0256" (8 holes)
- + 0.800mm / 0.0315" (84 holes)
- 0.900mm / 0.0354" (12 holes)
- ◇ 1.000mm / 0.0394" (10 holes)
- ⊠ 1.200mm / 0.0472" (12 holes)
- ✱ 1.250mm / 0.0492" (2 holes)
- ⊞ 1.300mm / 0.0512" (12 holes)
- ⊞ 1.400mm / 0.0551" (20 holes)

Figure 6 - PTH drill map



Drill Map:
× 3.200mm / 0.1260" (4 holes) (not plated)

Figure 7 - NPTH drill map

9. USEFUL LINKS

Git hub repo : <https://github.com/Mighty-Tronics/MightyDrive>